

# List of Presentations and Talks of Oliver Gutsche

Oliver Gutsche  
July 01, 2026

## 1. Scientific Computing and Scientific Software Infrastructure - High Performance Computing (HPC)

April 2025 — Presentation for the [Chicagoland Computational Traineeship in High Energy Particle Physics](#), [\(Material\)](#)

## 2. Scientific Computing and Scientific Software Infrastructure - Particle Physics Overview

April 2025 — Presentation for the [Chicagoland Computational Traineeship in High Energy Particle Physics](#), [\(Material\)](#)

## 3. Computing and Software Infrastructure

July 2024 — Presentation at the [Fermilab-CERN HCP Summer School 2024](#), [\(Material\)](#)

## 4. HL-LHC Computing

July 2023 — Presentation at the [USCMS Undergraduate Summer Internship 2023](#), [\(Material\)](#)

## 5. The U.S. CMS HL-LHC R&D Strategic Plan

May 2023 — Parallel Session Talk at the [26th International Conference on Computing in High Energy & Nuclear Physics \(CHEP2023\)](#), [\(Material\)](#)

## 6. Computing

September 2022 — Plenary talk given at the DOE/HEP Review of the Energy Frontier Laboratory Research Program, [\(Material available upon request\)](#)

## 7. Computing

August 2022 — Lecture given at [17th Hadron Collider Physics Summer School](#), [\(Material\)](#)

## 8. Exabytes and the Large Hadron Collider: the importance of computing excellence for scientific discovery

November 2019 — Seminar given at [Dr. Leonard Weisenthal Colloquium Series at Lewis University](#), [\(Material\)](#)

## 9. WLCG Web Proxy Auto Discovery for Dynamically Created Web Proxies

November 2019 — Parallel Session Talk at the [24th International Conference on Computing in High Energy & Nuclear Physics \(CHEP2019\)](#), [\(Material\)](#)

## 10. Bringing heterogeneity to the CMS software framework

November 2019 — Parallel Session Talk at the [24th International Conference on Computing in High Energy & Nuclear Physics \(CHEP2019\)](#), [\(Material\)](#)

## 11. Highlights from Parallel Session: Software & Computing

June 2019 — Plenary Talk at the [2019 US CMS Annual Collaboration Meeting](#), [\(Material\)](#)

## 12. Software & Computing

June 2019 — Plenary Talk at the [2019 US CMS Annual Collaboration Meeting](#), [\(Material\)](#)

## 13. CMS Software, Computing and Analysis Tools

January 2019 — Plenary Talk at the CMS Data Analysis School 2019 [\(CMSDAS 2019\)](#), [\(Material\)](#)

## 14. Exploring the Energy Frontier using Amazon Web Services

November 2018 — Amazon Booth at [Supercomputing 2018](#) in Dallas, TX, [\(Material\)](#)

## 15. The Future of Large Scale Scientific Computing - Trends in computing hardware

July 2018 — [US ATLAS Workshop 2018 Open Day](#) at the University of Pittsburgh, ([Material](#))

## **16. The Future of Large Scale Scientific Computing**

April 2018 — [Colloquium](#) given at the Physics Department at the University of Massachusetts Amherst, ([Material](#))

## **17. CMS Software, Computing and Analysis Tools**

January 2018 — Plenary Talk at the CMS Data Analysis School 2018 ([CMSDAS 2018](#)), ([Material](#))

## **18. Data Organisation, Management and Access (DOMA) in High Energy Physics**

November 2017 — Talk at the Workshop for Data Organisation, Management and Access (DOMA) in Astronomy, Genomics and High Energy Physics ([DOMA2017](#)), ([Material](#))

## **19. The Future of Large Scale Scientific Computing**

October 2017 — [Colloquium](#) given at the Department of Physics and Astronomy at Texas Tech University, ([Material](#))

## **20. Particle Physics - A world wide journey from recording particles to analysis using big computing**

September 2017 — Plenary Talk at the Chicago Council on Science & Technology Panel: [Fermilab and the New Frontiers of Physics](#), ([Material](#))

## **21. CMS Analysis and Data Reduction with Apache Spark**

August 2017 — Parallel Session Talk at the 18th International Workshop on Advanced Computing and Analysis Techniques in Physics Research ([ACAT 2017](#)), ([Material](#))

## **22. 50 years Fermilab - Computing Innovations**

June 2017 — Talk at the Fermilab 50th Anniversary [Symposium](#), ([Material](#))

## **23. Status and Plans of the CMS Big Data Project**

May 2017 — Talk at the [CERN Database Futures Workshop 2017](#), ([Material](#))

## **24. Data Management, Access and Organisation (DOMA)**

May 2017 — Summary Talk at the [2nd S2I2 HEP/CS Workshop](#), ([Material](#))

## **25. Data Analytics in Physics Data Reduction**

April 2017 — Talk at the [CERN openlab Workshop on Machine Learning and Data Analytics 2017](#), ([Material](#))

## **26. Panel Discussion on the S2I2 Conceptualization for HL-LHC**

March 2017 — Talk at the Openscience Grid All Hands Meeting ([OSG AHM 2017](#)), ([Material](#))

## **27. The Fermilab HEPCloud, or How to add 240 TFlops in an Hour or Two**

January 2017 — Talk at HEP Software Foundation [Workshop](#), ([Material](#))

## **28. Intel Big Data Project: CMS Physics Data Reduction Use Case**

December 2016 — Talk at the [CERN openlab Technical Workshop 2016](#), ([Material](#))

## **29. Big Data in HEP: a Comprehensive Use Case Study**

October 2016 — Parallel Session Talk at 22nd International Conference on Computing in High Energy and Nuclear Physics ([CHEP 2016](#)), ([Material](#))

## **30. Big Data at FNAL**

October 2016 — Plenary Talk given at the [8th INFIERI Workshop](#), ([Material](#))

## **31. Amazon and the Future of Computing**

October 2016 — [Ask-A-Scientist](#) Lecture given to the general public at Fermi National Accelerator Laboratory, ([Material](#))

## **32. Exascale and Exabytes: Future directions in HEP software and computing**

November 2015 — Seminar given at [Center of Research Computing Seminar Series](#) the University of Notre Dame, [\(Material\)](#)

### **33. Exascale and Exabytes - Computing at Fermilab for LHC and Intensity Frontier**

August 2015 — Seminar given at the University Los Andes in Bogota, Colombia, [\(Material\)](#)

### **34. Future directions in HEP Software and Computing**

August 2015 — Plenary talk given at the Meeting of the Division of Particles & Fields of the American Physical Society at the University of Michigan [\(DPF2015\)](#), [\(Material\)](#)

### **35. Diversity in Computing Technologies and Strategies for Dynamic Resource Allocation**

April 2015 — Plenary talk given at the Conference for Computing in High Energy Physics [\(CHEP2015\)](#), Okinawa, Japan , [\(Material\)](#)

### **36. CMS Computing Operations: Outlook on LHC Run 2**

March 2015 — Talk given at the Openscience Grid All Hands Meeting 2015 at Northwestern University [\(OSGAHM2015\)](#), [\(Material\)](#)

### **37. Computing Facilities and Services**

February 2015 — Talk given at the [Fermilab Institutional Review 2015](#), [\(Material\)](#)

### **38. Scientific Computing**

June 2014 — Plenary talk given at the [47th Annual Fermilab Users Meeting](#), [\(Material\)](#)

### **39. CMS computing operations: the year ahead**

April 2014 — Talk given at the Openscience Grid All Hands Meeting 2014 at SLAC [\(OSGAHM2014\)](#), [\(Material\)](#)

### **40. Recent Top Pair Asymmetry Measurements at CMS**

February 2014 — Seminar given at the [Fermilab Joint Experimental-Theoretical Physics Seminar](#), [\(Material\)](#)

### **41. Computing: From Data to Physics**

December 2013 — [Ask-A-Scientist](#) Lecture given to the general public at Fermi National Accelerator Laboratory, [\(Material\)](#)

### **42. Measurement of top quark properties with CMS**

November 2013 — Parallel session talk given at the 19th International Symposium on Particles, Strings, and Cosmology [\(PASCOS2013\)](#), Taipei, Taiwan, [\(Material\)](#)

### **43. CMS: T2 dynamic data placement and plans for T2 disk space handling**

November 2013 — Plenary talk given at the [2013 WLCG Collaboration Workshop](#) in Copenhagen, Denmark, [\(Material\)](#)

### **44. CMS: Disk/Tape Separation**

November 2013 — Plenary talk given at the [2013 WLCG Collaboration Workshop](#) in Copenhagen, Denmark, [\(Material\)](#)

### **45. Computing Support Facilities**

November 2013 — Talk given at the [2013 Fermi National Accelerator Laboratory Annual Science and Technology Review](#), [\(Material\)](#)

### **46. CMS computing operations in LHC run 1**

October 2013 — Parallel session talk given at the 2013 Conference for Computing in High Energy Physics [\(CHEP2013\)](#), Amsterdam, The Netherlands, [\(Material\)](#)

### **47. CMS Production Transfer Case Study**

August 2013 — Talk given at the [2013 Department of Energy ESNet Requirements Review](#), [\(Material\)](#)

### **48. Summary of CMS Operations Procedures**

May 2012 — Talk given at the [WLCG Collaboration Workshop 2012](#), ([Material](#))

#### **49. Summary of Event Processing Track**

October 2010 — Plenary talk given at the conference for Computing in High Energy and Nuclear Physics 2010 ([CHEP2010](#)) in Taipei (Taiwan), ([Material](#))

#### **50. Validation of software releases for CMS**

March 2009 — Parallel session talk given at the conference for Computing in High Energy and Nuclear Physics 2009 ([CHEP2009](#)) in Prague, Czech Republic, ([Material](#))

#### **51. Dilepton Final States Studies at CMS: From the Standard Model to New Physics**

January 2009 — Talk given at the [LHC and Dark Matter Workshop](#) at the University of Michigan in Ann Arbor, MI, ([Material](#))

#### **52. WLCG scale testing during CMS data challenges**

September 2007 — Parallel session talk given at the conference for Computing in High Energy and Nuclear Physics 2007 ([CHEP2007](#)) in Victoria, Canada, ([Material](#))

#### **53. Distributed CMS Analysis on the Open Science Grid**

February 2006 — Parallel session talk given at the conference for Computing in High Energy and Nuclear Physics 2006 ([CHEP2006](#)) in Mumbai, India, ([Material](#))